

Osteoarthritis

What causes arthritis?

The most common type of arthritis, called *osteoarthritis*, is caused by the breakdown of cartilage, the smooth material covering the ends of bones in your joints. Cartilage protects bones from rubbing and absorbs shock when you move. As arthritis worsens, the cartilage wears away, damaging the joint and causing pain. Your normally flexible joints become stiff when you have arthritis.

Joints may be prone to arthritis if they were injured in the past, if you're overweight, or if you have a family history of osteoarthritis. Although arthritis can develop at any age, you're more likely to develop it if you're older than age 60.

If you have arthritis, you may feel stiffness or an aching pain in the joint. You may feel worse when you get up in the morning or after you've been sitting a while. After moving around for about 30 minutes, you may feel less stiffness, but the pain doesn't go away.

How will my health care provider know I have arthritis?

Your health care provider will ask you about your symptoms, the joints that hurt, and the pattern of pain. She'll also examine the joint and order an X-ray of it. If your stiffness affects many joints or won't go away, she may order tests to rule out other problems, including other types of arthritis.

How will my health care provider treat my arthritis?

Although osteoarthritis can't be cured, you can learn to manage it with lifestyle changes, exercise, physical therapy, or medicine. (See *How does my arthritis medicine help me?*)

Other treatments that help some patients include the following:

- Glucosamine and chondroitin supplements may ease joint pain.
- Hyaluronic acid injected into a painful joint can lubricate and cushion it.



- Capsaicin cream rubbed into a joint may help ease pain too, although you may need to use it for several weeks before you feel relief.
- Steroids injected into a joint can relieve pain, but you shouldn't have more than three or four steroid injections in the same joint within 1 year.
- Put warm compresses on the joint before activity and use cold compresses on the joint afterward to reduce pain.

If a joint is badly damaged from arthritis, your health care provider may recommend arthroscopic surgery, joint fusion, or joint replacement.

What can I do to keep my arthritis from getting worse?

Make these tips part of your everyday life.

- **Keep moving.** When you're still, your joints will stiffen. Your primary care provider or physical therapist will show you exercises you should do every day to keep your joints flexible and your muscles strong. Strong, limber muscles help support your joints.
- **Get enough rest.** Pain is tiring. When you feel tired,

How does my arthritis medicine help me?

Find your arthritis medicine on this chart, then look across the table to find out how it helps you and things you should look out for. New research shows that taking NSAIDs or a COX-2 inhibitor increases the risk of a heart attack or stroke, so take any of these drugs only for a short period.

Drug	How it helps you	Things you should look out for
• Acetaminophen (Tylenol)	• Relieves minor arthritis pain	• Never take more than 4 grams/day (no more than 2.5 grams/day if you have liver problems or drink alcohol).
• Nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen (Motrin, Advil) or naproxen (Aleve)	• Relieves minor-to-moderate arthritis pain	• Some patients taking these drugs for a long period develop serious health problems, such as internal bleeding, high blood pressure, and worsening heart or kidney function.
• Cyclooxygenase-2 inhibitor; for example, celecoxib (Celebrex)	• Long-acting pain relief that's easier on the stomach than some other pain relief drugs	• Some patients taking this drug for a long period develop serious health problems, such as internal bleeding, high blood pressure, and worsening heart or kidney function. Don't take over-the-counter NSAIDs with this drug. Using this drug may increase the chance of a heart attack or stroke that can lead to death. It should not be used right before or after certain heart surgeries. Serious skin reactions or stomach problems, such as bleeding, can occur without warning and may cause death.
• Tramadol (Ultram)	• Relieves moderate-to-severe arthritis pain	• This drug isn't for you if you're at risk for seizures. Take a stool softener to prevent constipation.
• Tramadol/acetaminophen (Ultracet)	• Provides short-term pain relief	• Take a stool softener to prevent constipation.
• Opioids such as oxycodone (OxyContin SR)	• Provides long-acting pain relief for moderate to severe arthritis pain	• Take a stool softener to prevent constipation. Swallow the extended-release tablets whole. Don't participate in any activities that require mental alertness until you know how the drug affects you.

sit or lie down for 20 minutes.

- **Relax.** Relaxation isn't the same as rest. When you relax, you let go of the tension that builds up in your body. Tension can tighten muscles and cause pain.
- **Watch your weight.** Too much weight puts extra strain on your joints. Check with your health care provider to find out what weight is right for your height and body type.
- **Don't slouch.** Your mom was right. Good posture helps maintain healthy joints, from your neck all the way down to your knees.
- **Protect finger joints by using your hands correctly.**

If you need to lift something, try doing so with an open palm. Clenching your fingers around a heavy object strains the fingers. Your health care provider will explain how you can use good body mechanics when you lift or move objects.

- **Wear braces if necessary.** Wrist or knee braces support sore joints and help prevent more joint damage.
- **Be sensible.** If you can't handle a task because of your arthritis, ask for help.

With the right treatments and therapies, you can reduce pain and joint damage and continue to live an active life.

This patient-education guide has been adapted for the 5th-grade level using the Flesch-Kincaid and SMOG formulas. It may be photocopied for clinical use or adapted to meet your facility's requirements. Selected references are available upon request.

Special thanks to Tracy Kane, MEd, patient-education coordinator, Albert Einstein Health Care Network, Philadelphia, Pa.

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grams/day (no more than 2.5 grams/day if she has liver problems or drinks alcohol).

• **Nonsteroidal anti-inflammatory drugs (NSAIDs).** If acetaminophen doesn't manage symptoms, an NSAID, such as ibuprofen (Motrin, Advil) or naproxen (Aleve), may be recommended. At low doses, these drugs ease mild pain and relieve inflammation, and they relieve severe pain at higher doses. However, they're also associated with internal bleeding, high blood pressure, and worsening heart or kidney function when used at higher dosages for long periods. Another type of NSAID called a cyclooxygenase-2 (COX-2) inhibitor—for example, celecoxib (Celebrex)—may be safer for some patients for long-term use because it's less irritating to the gastrointestinal tract. However, the Food and Drug Administration (FDA) recently requested that manufacturers of marketed prescription NSAIDs such as celecoxib revise their package inserts to include a boxed warning and a medication guide highlighting the potential for increased risk of cardiovascular events, a serious risk of gastroin-

testinal bleeding, and serious skin reactions associated with these drugs. The warnings also include a contraindication for prescribing this class of drug to patients who've recently undergone or will undergo coronary artery graft bypass surgery.

The FDA has also asked all manufacturers of nonprescription NSAID products such as ibuprofen, naproxen, and ketoprofen (Orudis) to revise their labeling similarly. And it has asked manufacturers to include stronger reminders about limiting the dose and duration of treatment unless otherwise advised by a primary care provider, as well as a warning about potential skin reactions.

Be sure that patients taking NSAIDs of any kind use them as directed and inform their health care provider about any adverse reactions. They should also inform the prescribing health care provider if they have high blood pressure or kidney or liver problems.

• **Tramadol or tramadol/acetaminophen.** If your patient doesn't experience pain relief from acetaminophen and can't take an NSAID, her health care provider might prescribe tramadol (Ultram)

or tramadol/acetaminophen (Ultracet). These drugs aren't appropriate for patients at risk for seizures, however.

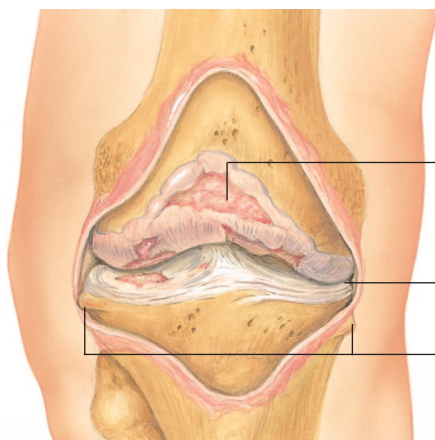
• **Long-acting opioids.** Opioids, such as oxycodone (OxyContin SR) or a fentanyl transdermal patch (Duragesic), may be indicated for severe osteoarthritis pain. If your patient is taking one of these drugs, advise her to use a stool softener to prevent constipation.

The other arthritis

Rheumatoid arthritis, which is less common than osteoarthritis, nevertheless affects more than 2 million people in the United States, two-thirds of them women. Most people who have rheumatoid arthritis develop it in their 50s and 60s, but some experience symptoms in their 20s or 30s, or even younger.

Unlike osteoarthritis, rheumatoid arthritis is an autoimmune disease that attacks the synovium (joint lining). Some specialists believe that a genetic predisposition to the disease is the catalyst; others believe that bacteria or viruses may trigger rheumatoid arthritis. Regardless of the cause, the attack causes synovial

Knee with osteoarthritis



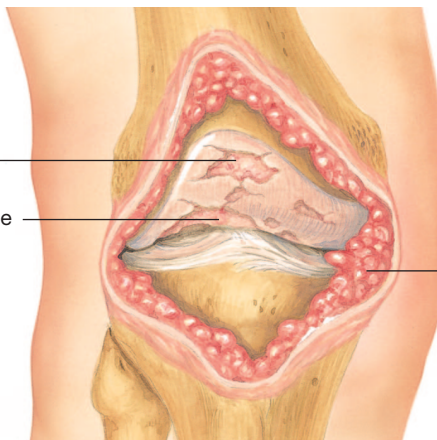
Erosion of
cartilage

Erosion of bone

Joint space
narrowing

Osteophyte

Knee with rheumatoid arthritis



Pannus covers
synovial
membrane

inflammation and damages bone and cartilage.

Like osteoarthritis, rheumatoid arthritis causes joint pain and stiffness. But the pain and stiffness of rheumatoid arthritis are usually more widespread than that of osteoarthritis, and they affect joints on both sides of the body. The stiffness of osteoarthritis typically eases with activity; that of rheumatoid arthritis doesn't.

Rheumatoid arthritis pain can be chronic and persistent, or it can occur in waves. Patients may also develop nodules at pressure points, such as behind the elbows and in the proximal interphalangeal and metacarpal phalangeal joints of the hands. Some patients may develop other manifestations of autoimmune or inflammatory disease, such as interstitial lung disease, pleurisy, pericarditis, keratitis sicca, or rheumatoid vasculitis.

Rheumatoid arthritis can cause additional, more general signs of ill health, such as a persistent low fever, loss of appetite, lethargy, and anemia.

Diagnosis and treatment

A rheumatologist will diagnose rheumatoid arthritis based on the patient's symptoms, pain pattern, X-rays (to determine bone and cartilage breakdown), and blood tests. The blood tests will include a rheumatoid factor (RF) blood test or tests of other inflammatory markers, such as erythrocyte sedimentation rate and C-reactive protein level. However, a positive test for RF doesn't definitively indicate rheumatoid arthritis; it may also point to other conditions. More than one-half of patients with high levels of RF have Sjögren's syndrome, a systemic inflammatory disorder that affects the mucous membranes, causing dry mouth, decreased tear production, and other

dry conditions of the body's membranes. If a patient's RF test is positive but she doesn't have rheumatoid arthritis or Sjögren's syndrome, the results may indicate another condition, including endocarditis; systemic lupus erythematosus; tuberculosis; syphilis; sarcoidosis; cancer; viral infection; or disease of the liver, lung, or kidney.

Likewise, a negative finding doesn't necessarily rule out rheumatoid arthritis: Patients can have a mild form of the condition and still test negative for RF. A new blood test specific for rheumatoid arthritis,

For your patient with rheumatoid arthritis, the keys to managing the disease are balancing exercise and rest and taking medications as prescribed to reduce inflammation and relieve pain.

the anticyclic citrullinated peptide antibody test, may also be used to help make a diagnosis. In one study, the test was shown to identify more than 70% of cases of early rheumatoid arthritis with greater than 98% specificity.

Because irreversible damage takes place in the first 2 years of rheumatoid arthritis, early diagnosis and intervention are crucial. The goal of treatment is to control inflammation, prevent joint damage, and relieve pain.

The key to managing rheumatoid arthritis is to balance exercise and rest. When the joints are inflamed and the disease is active, patients should rest as much as possible. When the disease is under control, patients should exercise to strengthen muscles and maintain range of motion.

Medications are a mainstay of

treating this disease. The patient's rheumatologist will prescribe disease-modifying antirheumatic drugs, which are used singly or in combination. These drugs include leflunomide (Arava), methotrexate sodium (Rheumatrex), sulfasalazine (Azulfidine), hydroxychloroquine (Plaquenil), azathioprine (Imuran), cyclosporine, and gold compounds. Onset of action takes 3 to 12 weeks, so NSAIDs or glucocorticoids may be prescribed until the antirheumatic drugs take effect.

For a patient whose arthritis isn't adequately controlled, the primary

care provider may order a biologic response modifier, such as etanercept (Enbrel), adalimumab (Humira), or infliximab (Remicade). These drugs inhibit cytokines (proteins released by cells that direct actions of other cells), specifically, tumor necrosis factor. These immune system components promote inflammation.

Adalimumab, the newest drug for moderate-to-severe rheumatoid arthritis, can be administered once every other week; etanercept needs to be given weekly. Both of these drugs are administered subcutaneously. Patients may self-administer Adalimumab. It's supplied in pre-filled syringes to make it easier for a person who is disabled by arthritis. Infliximab is administered via intravenous infusion at specific intervals. Recent research has shown that combining etanercept with methotrexate can induce clinical remission

and stop joint damage in some patients with rheumatoid arthritis.

Another, less commonly used option is Prosorba Column therapy. For this treatment, the patient's blood is drawn and pumped into an apheresis machine, which separates plasma from blood cells. The plasma passes through the Prosorba Column—a plastic cylinder that contains a sandy substance coated with protein A, which binds rheumatoid arthritis antibodies. The plasma is then reunited with blood cells and the blood is returned to the patient's body through a vein in the arm opposite the one from which it was drawn. The procedure treats 4 to 8 cups of plasma and takes about 2 hours. The body's immune response is reduced because antibodies have been removed from the blood.

Patients with severe joint damage from either osteoarthritis or rheumatoid arthritis may be candidates for arthroscopic surgery, joint fusion, or joint replacement surgery.

Living well

Living with arthritis isn't always easy, but patients can do a lot to keep the disease from interfering with daily life. See the patient education reproducible for tips that can help patients cope with osteoarthritis. With the right treatments and therapies, your patient can minimize pain and joint damage and continue to live an active life. **LPN**

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On the Web

Arthritis Foundation Disease Center: <http://www.arthritisfoundation.org/conditions/diseasecenter>

American Academy of Orthopaedic Surgeons:
<http://orthoinfo.aaos.org>

National Institute of Arthritis and Musculoskeletal and Skin Diseases:
<http://www.niams.nih.gov>

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